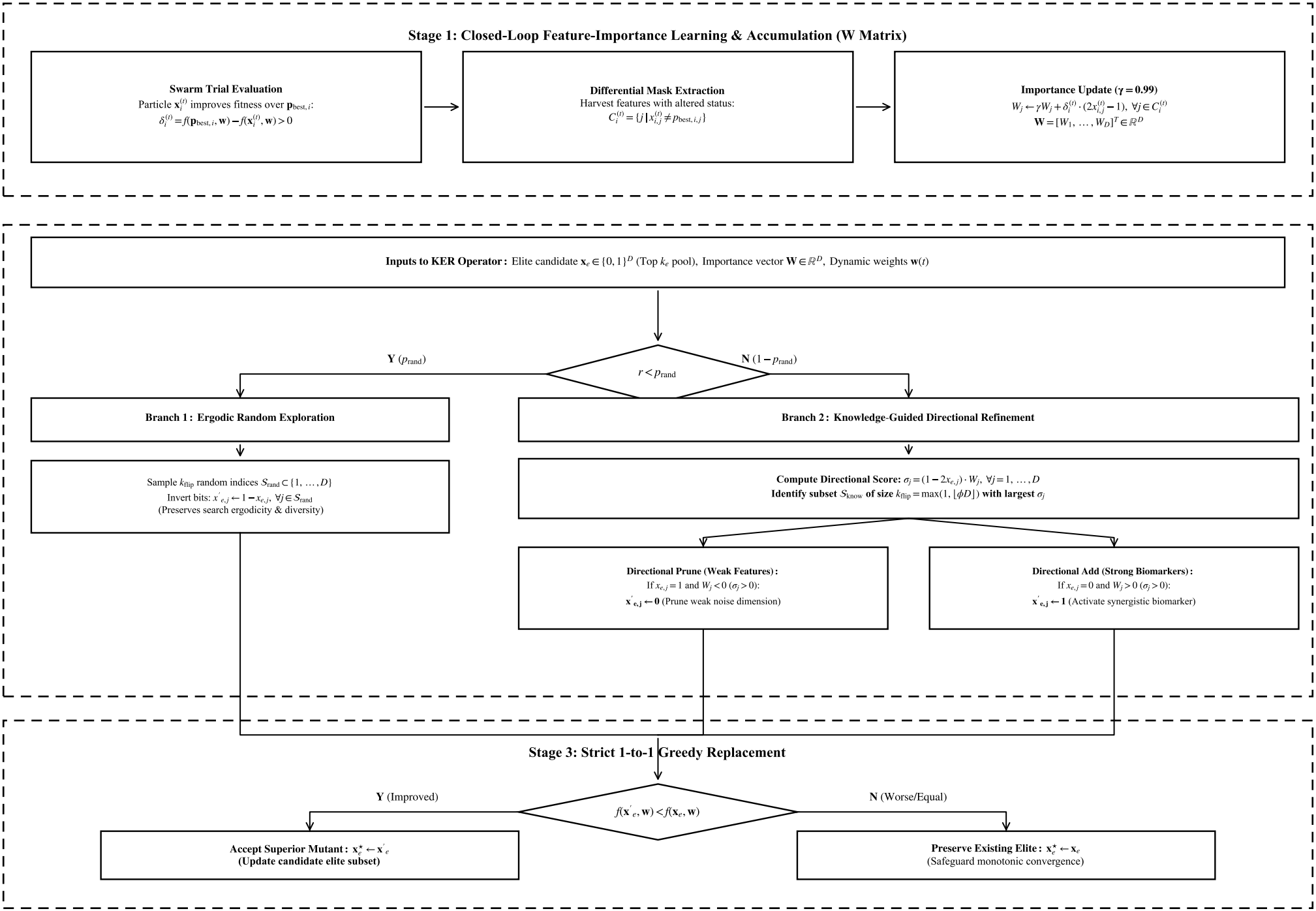

Stage 1: Closed-Loop Feature-Importance Learning & Accumulation (W Matrix)
Swarm Trial Evaluation
Particle $\mathbf{x}_i^{(t)}$ improves fitness over $\mathbf{p}_{\text{best},i}$:
$\delta_i^{(t)} = f(\mathbf{p}_{\text{best},i}, \mathbf{w}) - f(\mathbf{x}_i^{(t)}, \mathbf{w}) > 0$
Differential Mask Extraction
Harvest features with altered status:
$C_i^{(t)} = \{j \mid x_{i,j}^{(t)} \neq p_{\text{best},i,j}\}$
Importance Update ($\gamma = 0.99$)
$W_j \leftarrow \gamma W_j + \delta_i^{(t)} \cdot (2x_{i,j}^{(t)} - 1),\ \forall j \in C_i^{(t)}$
$\mathbf{W} = [W_1, \ldots, W_D]^T \in \mathbb{R}^D$
Inputs to KER Operator: Elite candidate $\mathbf{x}_e \in \{0,1\}^D$ (Top k_e pool), Importance vector $\mathbf{W} \in \mathbb{R}^D$, Dynamic weights $\mathbf{w}(t)$
$r < p_{\text{rand}}$
Y (p_{rand})
N ($1 - p_{\text{rand}}$)
Branch 1: Ergodic Random Exploration
Sample k_{flip} random indices $\mathcal{S}_{\text{rand}} \subset \{1, \ldots, D\}$
Invert bits: $x'_{e,j} \leftarrow 1 - x_{e,j},\ \forall j \in \mathcal{S}_{\text{rand}}$
(Preserves search ergodicity & diversity)
Branch 2: Knowledge-Guided Directional Refinement
Compute Directional Score: $\sigma_j = (1 - 2x_{e,j}) \cdot W_j,\ \forall j = 1, \ldots, D$
Identify subset $\mathcal{S}_{\text{know}}$ of size $k_{\text{flip}} = \max(1, \lfloor \phi D \rfloor)$ with largest σ_j
Directional Prune (Weak Features):
If $x_{e,j} = 1$ and $W_j < 0$ ($\sigma_j > 0$):
$\mathbf{x'_{e,j}} \leftarrow \mathbf{0}$ (Prune weak noise dimension)
Directional Add (Strong Biomarkers):
If $x_{e,j} = 0$ and $W_j > 0$ ($\sigma_j > 0$):
$\mathbf{x'_{e,j}} \leftarrow \mathbf{1}$ (Activate synergistic biomarker)
Stage 3: Strict 1-to-1 Greedy Replacement
$f(\mathbf{x}'_e, \mathbf{w}) < f(\mathbf{x}_e, \mathbf{w})$
Y (Improved)
N (Worse/Equal)
Accept Superior Mutant: $\mathbf{x}_e^\star \leftarrow \mathbf{x}'_e$
(Update candidate elite subset)
Preserve Existing Elite: $\mathbf{x}_e^\star \leftarrow \mathbf{x}_e$
(Safeguard monotonic convergence)